
Design of Intelligent Edge and Time-Series Data Flow Temperature Control System

Разработка интеллектуальной системы управления температурой
на основе периферийных вычислений и потоковой обработки временных рядов

Student / Студент: Wang Gen

Руководитель: В. С. Разумейчик

Department / Кафедра: «ЭВМ и систем»

BrSTU / БрГТУ

2026

Objective And Implemented Scope

Objective

Build an end-to-end intelligent temperature-control prototype from edge control to data-driven optimization.

The thesis uses the word design, but the work includes implementation, integration, and prototype validation.

Implemented Scope

- 1 Edge closed-loop control**
Sensor, PID-compatible control, PWM output, local safety
- 2 MQTT command and telemetry**
Device reporting, parameter downlink, edge ACK
- 3 Reactive Data Hub**
Ingestion, validation, filtering, aggregation, alarms
- 4 HMI operation layer**
Monitoring, history, alarms, parameter management
- 5 AI-assisted optimization**
Feature extraction, candidate ranking, preview, evaluation

Requirements And Design Goals

Functional Requirements

- Temperature acquisition and PWM control
- Telemetry upload, storage, and history query
- Alarm, parameter, and device management
- AI recommendation with preview and confirmation
- Command downlink with edge acknowledgement

Design Goals

Safety

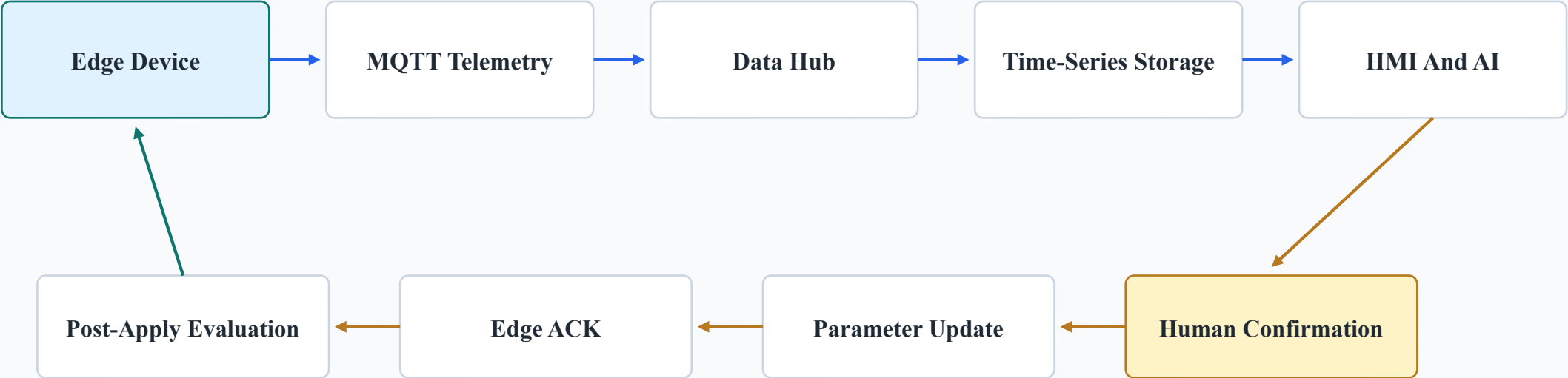
Observability

Traceability

Maintainability

Real-time control, data flow, operation, and decision support are separated into clear layers.

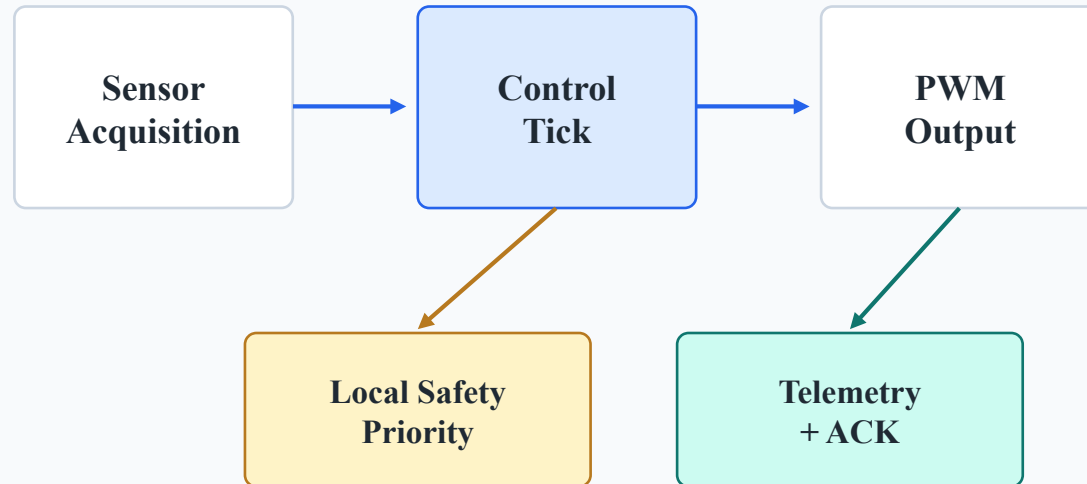
End-to-End Closed Loop Architecture



Key Principles

- Real-time safety remains on the edge device.
- AI recommends parameters; humans confirm before applying.

Edge Control, Hardware Safety And Firmware



Edge Responsibilities

- ESP32-WROOM-32 with DS18B20 sensing
- GPIO18 PWM drives NMOS3400 heater switch
- Thermal switch cuts heater power on over-temperature
- PID-compatible control with limiting and anti-windup
- Fault latch: invalid sensor or over 65 C forces PWM off
- Runtime parameters saved in ESP32 NVS

Telemetry And Control Metrics

Telemetry

- Temperature, setpoint, and error
- PWM duty and saturation state
- PID parameters and control mode
- Sensor validity and fault latch
- Timestamp, device ID, and run ID

Control-quality
features

Control Metrics

- Mean absolute error
- Overshoot
- Settling time
- In-band ratio
- Temperature swing
- Saturation ratio

Telemetry is used for display, control-quality evaluation, and AI feature extraction.

Reactive Data Hub And Time-Series Flow



Processing Responsibilities

- HiveMQ async MQTT consumer
- Reactor pipeline with backpressure
- Device-lane ordered processing
- Typed validation before persistence

Storage And Observation

- Steady-state telemetry filtering
- Summary aggregation for skipped windows
- Rule-state alarm facts
- Runtime throughput and failure metrics

AI-Assisted PID Recommendation

AI Role

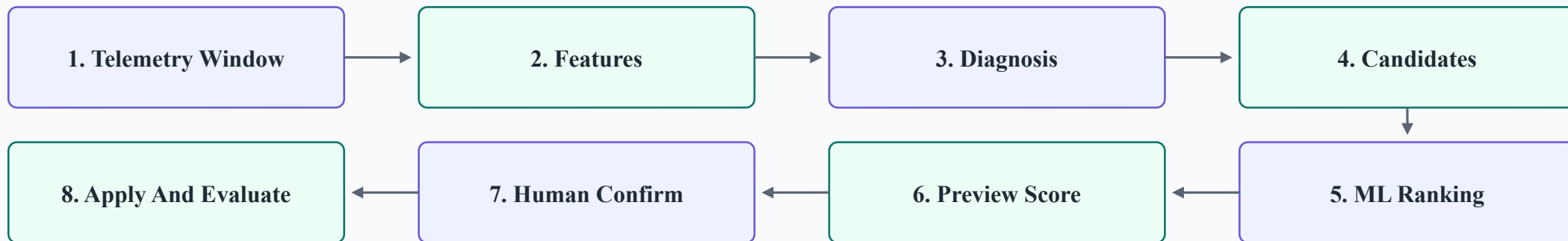
- Decision support, not direct actuation
- Real-time PID remains on edge
- Human confirmation before downlink

Rule-Based Layer

- Window feature extraction
- Control problem diagnosis
- Conservative PID candidates

ML And Preview Layer

- joblib models rank candidates
- Preview simulates PID response
- Score combines success and risk



$$\text{total_score} = 0.65 \times \text{success_score} + 0.35 \times \text{gap_score}$$

Deployment, Pressure Test And Operations

Deployment

- Edge firmware or simulator
- Docker Compose for software services
- MQTT broker and Java Data Hub
- TDengine time-series storage
- PostgreSQL, HMI, and AI services

Validation And Operation

- End-to-end telemetry and history
- Parameter set and ACK workflow
- Data Hub pressure baseline
- Metrics for drops and failures
- Database maintenance scripts

Results, Limitations And Future Work

Main Results

- End-to-end prototype implemented
- Edge control, safety, telemetry, and ACK verified
- Data Hub filtering, aggregation, storage verified
- HMI operation and AI workflow verified

Limitations

- Prototype-level thermal validation
- Physical load experiments should be extended
- Long-term quantitative datasets are still limited

Future Work

- More hardware protection and enclosure refinement
- Broader device and scenario testing
- Stronger model lifecycle and adaptive tuning
- Deployment automation and monitoring

The work combines system design, implementation, and prototype validation.